

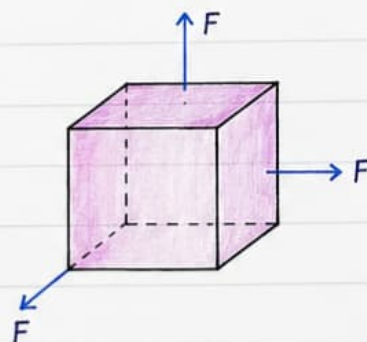
Chapter 8

Mechanical Properties of Solids

[NCERT Based Notes]

INTRODUCTION

Solids are rigid and resist change in shape or size when external forces are applied on them. However, under sufficient force, they undergo **deformation**. The study of deformation and related properties of solids is called **mechanical properties of solids**.

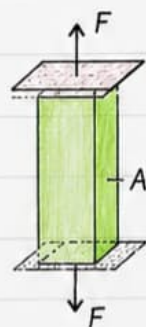


1. STRESS (σ)

- It is the internal restoring force per unit area developed in a body when external force is applied.
- Mathematically, stress (σ) = $\frac{\text{Internal force (F)}}{\text{Area (A)}}$

$$\sigma = \frac{F}{A}$$

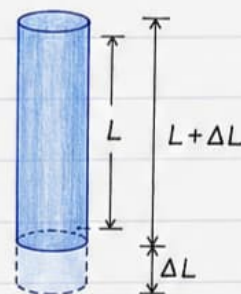
SI Unit: Pascal (Pa) = N/m^2



2. STRAIN (ϵ)

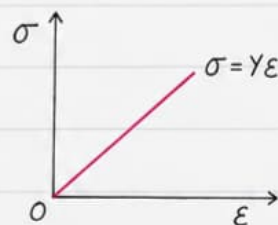
- It is the fractional change in dimension of the body produced by the applied stress.
- Mathematically, strain (ϵ) = $\frac{\text{Change in length } (\Delta L)}{\text{Original length } (L)}$
- Strain is dimensionless (no unit).

$$\epsilon = \frac{\Delta L}{L}$$



3. HOOKE'S LAW

- Within the elastic limit, stress is directly proportional to strain.
- Mathematically, $\sigma \propto \epsilon \Rightarrow \sigma = Y\epsilon$
- Here, Y is called Young's modulus or modulus of elasticity.
- SI Unit of Y: Pascal (Pa)



4. TYPES OF DEFORMATION

(i) Longitudinal Deformation

Change in length along the direction of applied force. (Tensile or Compressional)

(ii) Volumetric Deformation

Change in volume of the body due to uniform stress on all sides.

(iii) Shear Deformation

Change in shape of the body at constant volume.

(iv) Torsional Deformation

Twisting of the body about its axis when a couple is applied.

5. BULK MODULUS (K)

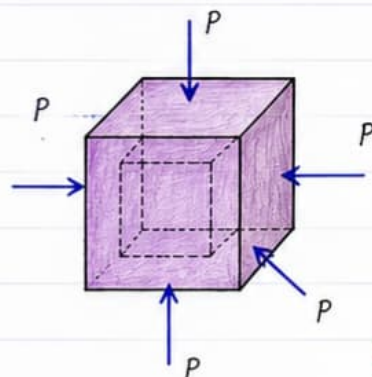
- It is the ratio of increase in pressure (P) to the corresponding fractional decrease in volume.
- It measures the resistance of a solid to uniform compression.
- Mathematically,

$$K = \frac{\text{Increase in pressure (P)}}{\text{Fractional decrease in volume}}$$

$$= - \frac{P}{\frac{\Delta V}{V}} = - \frac{PV}{\Delta V}$$

SI Unit : Pascal (Pa) = N/m^2

Uniform compression



V = Original volume

ΔV = Decrease in volume

K is always positive.

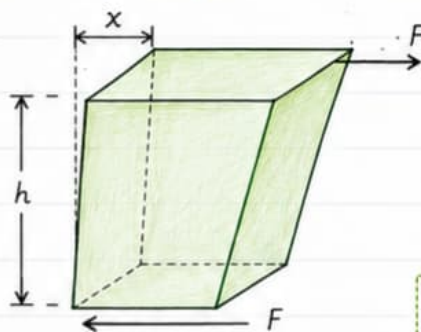
6. SHEAR MODULUS (η)

- It is the ratio of shear stress (τ) to the shear strain.
- It measures the resistance of a solid to shear deformation.
- Mathematically,

$$\eta = \frac{\text{Shear stress } (\tau)}{\text{Shear strain}} = \frac{\tau}{\text{Shear strain}}$$

SI Unit : Pascal (Pa) = N/m^2

Shear deformation



Shear strain (angular strain)

$$\gamma = \frac{x}{h}$$

(dimensionless)

$$\tau = \frac{F}{A}$$

(Shear stress)

7. MODULUS OF RIGIDITY (η) – RELATION WITH γ AND K

- For an isotropic, homogeneous and perfectly elastic solid, the three elastic moduli are related by:

$$\eta = \frac{\gamma}{2(1+\sigma)} = \frac{3K(1-2\sigma)}{2(1+\sigma)}$$

where, σ = Poisson's ratio

Typical Values (Approx.)

Material	γ (Pa)	K (Pa)	η (Pa)
Steel	2.0×10^{11}	1.6×10^{11}	8.0×10^{10}
Aluminium	7.0×10^{10}	7.0×10^{10}	2.6×10^{10}
Copper	1.1×10^{11}	1.4×10^{11}	4.2×10^{10}
Rubber	1.0×10^6	2.0×10^6	3.0×10^5

8. ENERGY STORED IN ELASTIC DEFORMATION

- When a body is deformed elastically, some potential energy is stored in it.
- (i) In case of stretching (or compression):

$$\text{Work done} = \frac{1}{2} F (\Delta L)$$

$$\text{Energy stored} = \frac{1}{2} (\text{stress}) \times (\text{strain}) \times \text{Volume}$$

- (ii) In case of shear deformation:

$$\text{Energy stored} = \frac{1}{2} (\text{shear stress}) \times (\text{shear strain}) \times \text{Volume}$$

Key Points ☆

- ✓ Solids resist deformation due to their intermolecular forces.
- ✓ Stress produces strain and both are proportional within elastic limit.
- ✓ Elastic behaviour is temporary and reversible.
- ✓ Different materials have different elastic moduli.
- ✓ Moduli depend on temperature. (Generally decrease with increase in temperature)

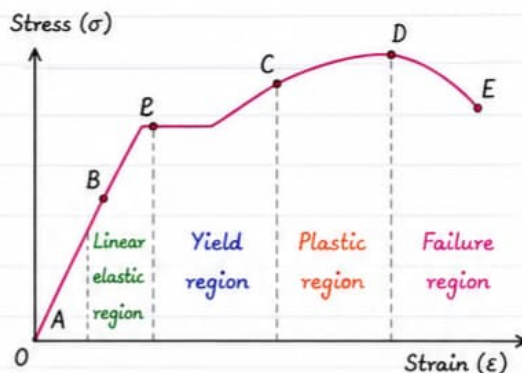


Higher the elastic moduli, stiffer is the material. ☆



9. STRESS-STRAIN CURVES

- Stress (σ) is plotted along y-axis and strain (ϵ) along x-axis.
- Different regions of the curve for a ductile material:



OA : Proportional limit
 AB : Elastic limit
 B : Yield point
 BC : Yield plateau
 C : Ultimate stress
 D : Necking begins
 E : Breaking point

✦ Linear elastic region (O-A):

Stress is directly proportional to strain.
 (Hooke's law is obeyed)

✦ Elastic region (A-B):

Material regains its original shape on removal of stress.

✦ Yield region (B-C):

Large strain occurs with little or no increase in stress.

✦ Plastic region (C-D):

Permanent deformation occurs.

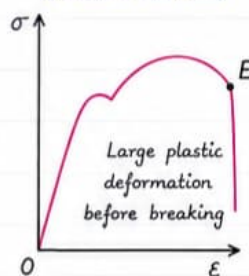
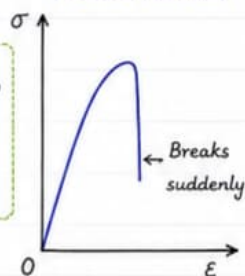
✦ Failure region (D-E):

Necking occurs and stress decreases till the material breaks.

Brittle vs Ductile

(Brittle Material)

(Ductile Material)



IMPORTANT

Brittle materials do not show yield region. They break suddenly with very little plastic deformation. ☆



APPLICATIONS



High Young's Modulus



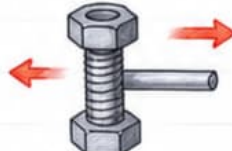
Steel is used in building bridges and structures as it is very stiff.

High Bulk Modulus



Materials with high bulk modulus are used in deep-sea vessels.

High Shear Modulus



Used in making nuts, bolts and machine parts to resist shear forces.

Torsional Deformation



Shafts and drill rods are designed to withstand twisting forces.

10. SOLVED EXAMPLES

Example 1: A steel wire of length 2.0 m and radius 0.5 mm is stretched by a load of 500 N. If Y (Young's modulus of steel) = 2.0×10^{11} Pa, find the increase in length.

Solution: Given,

$$L = 2.0 \text{ m},$$

$$r = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$$

$$F = 500 \text{ N}, Y = 2.0 \times 10^{11} \text{ Pa}$$

$$\text{Increase in length, } \Delta L = ?$$

Formula:

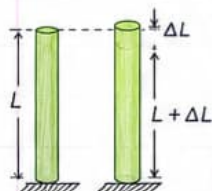
$$Y = \frac{F/A}{\Delta L/L}$$

$$\Rightarrow \Delta L = \frac{FL}{AY}$$

$$\text{Area, } A = \pi r^2 = \pi (0.5 \times 10^{-3})^2 = 7.85 \times 10^{-7} \text{ m}^2$$

$$\Delta L = \frac{FL}{AY} = \frac{500 \times 2.0}{7.85 \times 10^{-7} \times 2.0 \times 10^{11}} \\ = 1.27 \times 10^{-3} \text{ m} = 1.27 \text{ mm}$$

Increase in length = 1.27 mm ☆



Example 2: A cube of side 10 cm is subjected to a uniform pressure of 2.0×10^7 Pa. If the bulk modulus of the material is 5.0×10^{10} Pa, find the decrease in volume.

Solution: Given,

$$\text{Side of cube, } a = 10 \text{ cm} = 0.1 \text{ m}$$

$$\text{Pressure, } P = 2.0 \times 10^7 \text{ Pa}$$

$$\text{Bulk modulus, } K = 5.0 \times 10^{10} \text{ Pa}$$

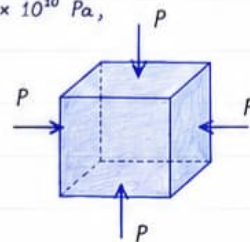
$$\text{Decrease in volume, } \Delta V = ?$$

$$K = \frac{P}{\frac{\Delta V}{V}} \Rightarrow \Delta V = \frac{PV}{K}$$

$$\text{Original volume, } V = a^3 = (0.1)^3 = 1.0 \times 10^{-3} \text{ m}^3$$

$$\Delta V = \frac{2.0 \times 10^7 \times 1.0 \times 10^{-3}}{5.0 \times 10^{10}} \\ = 4.0 \times 10^{-7} \text{ m}^3$$

Decrease in volume = $4.0 \times 10^{-7} \text{ m}^3$ ☆



Note: Moduli (Y , K , η) have the same SI unit $\rightarrow$ Pascal (Pa) = N/m^2



11. FACTORS AFFECTING ELASTIC PROPERTIES

- **Nature of material** : Different materials have different values of Y , K and η .
- **Temperature** : Elastic moduli generally **decrease** with increase in temperature.
- **Impurities / defects** : Presence of impurities or defects reduces the elastic behaviour of solids.

Remember !

- ✓ Stress → Internal restoring force / Area
- ✓ Strain → Change in dimension / Original dimension
- ✓ Within elastic limit → Body returns to original shape.
- ✓ Beyond elastic limit → Permanent deformation occurs.

12. LIMITATIONS OF ELASTICITY

- Every material behaves elastically only up to a certain limit.
- If stress exceeds elastic limit, material shows permanent deformation and does not regain its original shape.
- Brittle materials have very small elastic limit whereas ductile materials have large elastic limit.

Ductile vs Brittle

Ductile (e.g., Iron)

- Large elastic limit
- Shows large plastic deformation
- Does not break suddenly

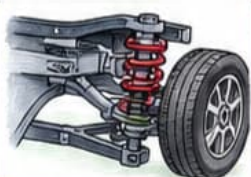
Brittle (e.g., Glass)

- Small elastic limit
- Negligible plastic deformation
- Breaks suddenly with no warning

13. ELASTIC MODULI OF COMMON MATERIALS (Approx.)

Material	Young's Modulus (Y) (Pa)	Bulk Modulus (K) (Pa)	Shear Modulus (η) (Pa)	Nature
Steel	2.0×10^{11}	1.6×10^{11}	8.0×10^{10}	Ductile
Aluminium	7.0×10^{10}	7.0×10^{10}	2.6×10^{10}	Ductile
Copper	1.1×10^{11}	1.4×10^{11}	4.2×10^{10}	Ductile
Glass	6.0×10^{10}	3.6×10^{10}	2.5×10^{10}	Brittle
Rubber	1.0×10^6	2.0×10^6	3.0×10^5	Highly elastic

14. REAL LIFE EXAMPLES



(i) Springs in vehicles

Made of steel having high Young's modulus so that they can withstand large stresses and return to original shape.



(ii) Shock absorbers

Use materials with suitable elastic properties to absorb shocks and reduce sudden jerks.



(iii) Bridges and buildings

Constructed with materials having high Y to resist deformation due to heavy loads and own weight.



(iv) Rubber bands

Rubber has very low Y and η , so it can be stretched easily and returns to original shape.

15. SUMMARY

- ✓ Solids resist deformation and exert restoring forces when external forces are applied.
- ✓ Stress is internal restoring force per unit area.
- ✓ Strain is fractional change in dimensions.
- ✓ Hooke's law holds good only within elastic limit.
- ✓ Three elastic moduli : Y (tension/compression), K (bulk), η (shear) – measure resistance to different types of deformation.
- ✓ Stress-strain curve shows different regions : linear, yield, plastic and failure.
- ✓ Energy is stored in a body during elastic deformation.
- ✓ Elastic moduli depend on material, temperature and impurities.

One Line Takeaway

The ability of a solid to resist deformation and return to its original shape is described by its mechanical properties.